

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- AERSURFACE
- Expectation
 - Understand the role of AERSURFACE and be able to use it
- Readings: AERSURFACE User guides

Components of AERMOD

- AERMET: meteorological data preprocessor
 - AERSURFACE: for surface characteristics
 - AERMINUTE: for 1-min ASOS data
- AERMAP: terrain data preprocessor
- AERMOD: uses input data (meteorological, terrain data etc) to perform refined analysis
 - BPIPPRIM: a multi-building dimensions program incorporating the GEP technical procedures for building downwash applications.

AERSURFACE

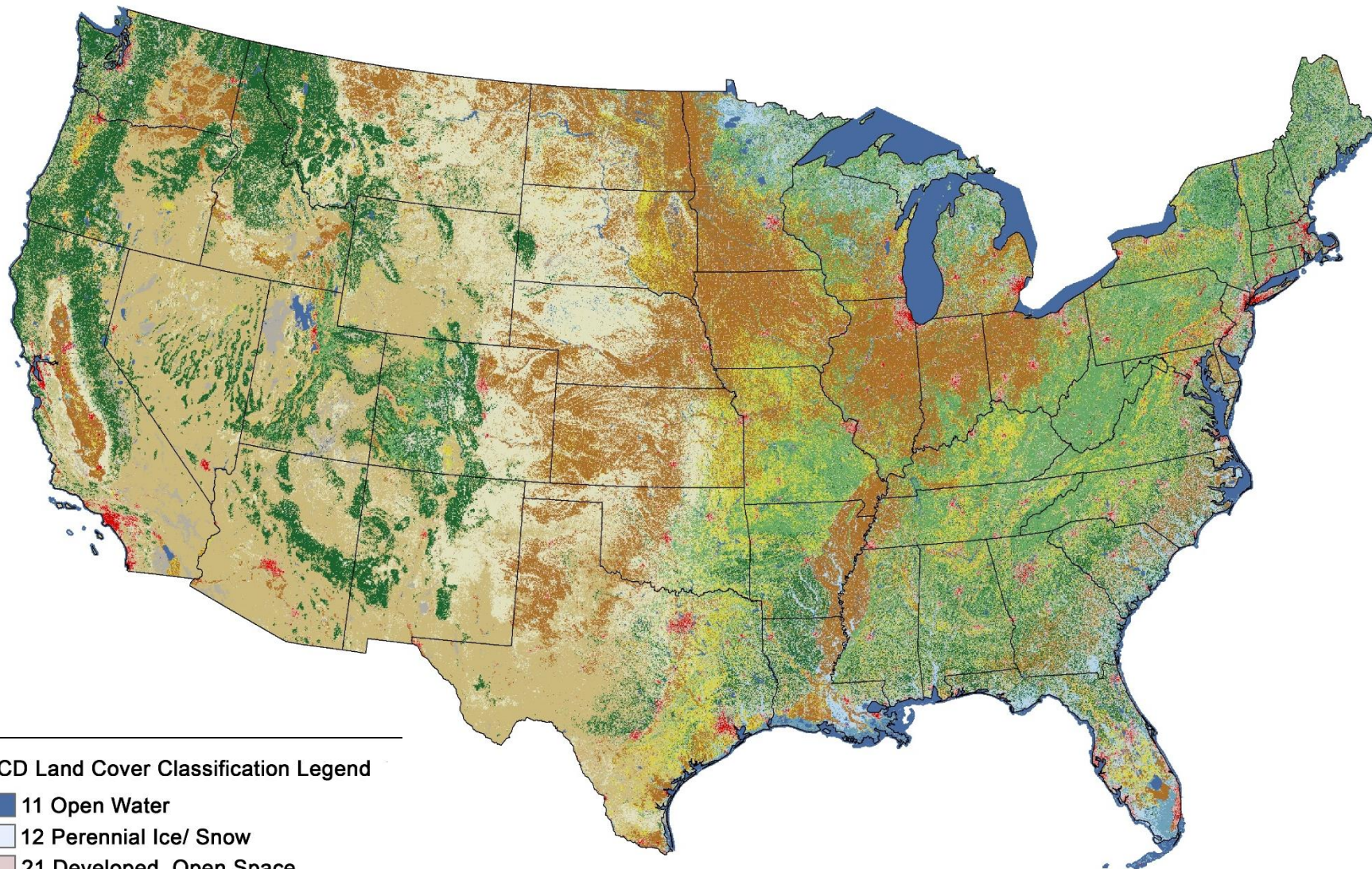
- AERSURFACE helps develop surface characteristics for AERMOD
 - Noon-time albedo
 - % sunlight reflected back, normally ranges from 0.1 – 0.9
 - Daytime Bowen ratio
 - How much energy were used to increase temperature vs used to evaporate water, normally ranges from 0.1-10
 - Surface roughness length
 - Lowest height at which wind speed is zero
 - Over which wind increases according to a power law
 - < 1 mm in very flat surface, a few meter for urban region
 - AERMOD most sensitivity to roughness length among them

General procedures

- Reads National Land Cover Database (NLCD)
 - Customized land cover data is possible
- Assign typical surface characteristic to each land use type near the chosen site
- Calculate averaged three surface characteristics
 - Averaged value for albedo and Bowen ratio
 - Spatially varying roughness length possible
 - By direction (up to 16 sectors, AERMOD only accept 12)
 - By time (annual, seasonal, monthly)

National Land Cover Database

- Land cover type map for entire US
 - Currently 20 different types of land cover
 - Open water, developed, shrub, forest etc
- 30 m resolution, mainly based on Landsat satellite data
- Used in numerous applications
- Has percent impervious and tree canopy data
 - % of surface that water can't seep through
 - % of surface covered by tree canopy
- Check data through MRLC Consortium Viewer
 - <https://www.mrlc.gov/viewerjs/>
- More on how Landsat works
 - https://www.youtube.com/watch?v=YP0et8I_bvY



NLCD Land Cover Classification Legend

- 11 Open Water
- 12 Perennial Ice/ Snow
- 21 Developed, Open Space
- 22 Developed, Low Intensity
- 23 Developed, Medium Intensity
- 24 Developed, High Intensity
- 31 Barren Land (Rock/Sand/Clay)
- 41 Deciduous Forest
- 42 Evergreen Forest
- 43 Mixed Forest

- 51 Dwarf Scrub*
- 52 Shrub/Scrub
- 71 Grassland/Herbaceous
- 72 Sedge/Herbaceous*
- 73 Lichens*
- 74 Moss*

- 81 Pasture/Hay
- 82 Cultivated Crops
- 90 Woody Wetlands
- 95 Emergent Herbaceous Wetlands

* Alaska only

AERSURFACE

- AERSURFACE not mandated regulatory option
 - Partially because NLCD data not that accurate for some small region and they keeps changing
 - AERMOD applies only to short range
- Previous AERSURFACE version can be run interactively or via input file
- Latest version can only be run via input file

AERSURFACE

- Latest AERSURFACE uses similar control files as in AERMOD
- The control file contains pathways and keywords
- Pathways are only two letters
 - CO: overall job **C**ontrol
 - OU: **O**Utput information
- Comment lines start with “**”

CO pathway

AERSURFACE CO pathway

- Start and end CO pathway
 - CO STARTING
 - CO FINISHED
- Between start and finish
 - CO TITLEONE *a bunch of title info*
 - Information to identify this AERMAP run
 - CO TITLETWO *more title info*
 - More information if needed, optional
- Can ignore “CO ” when between STARTING and FINISHED
 - Line should start with three empty spaces

AERSURFACE CO pathway

- CO OPTIONS

- Optional

- Choose PRIMARY or SECONDARY

- Recall AERMET needs two sets of data when both on-site and NWS data used

- Choose ZORAD or ZOEFF

- ZORAD: Default, inverse distance weighted surface roughness within given distance (1 km default)

- CO ZORADIUS set default radius for ZORAD

- ZOEFF: Estimate distance based on existing conditions

- CO ANEM_HGT set anemometer height

- E.g: CO OPTIONS PRIMARY ZORAD

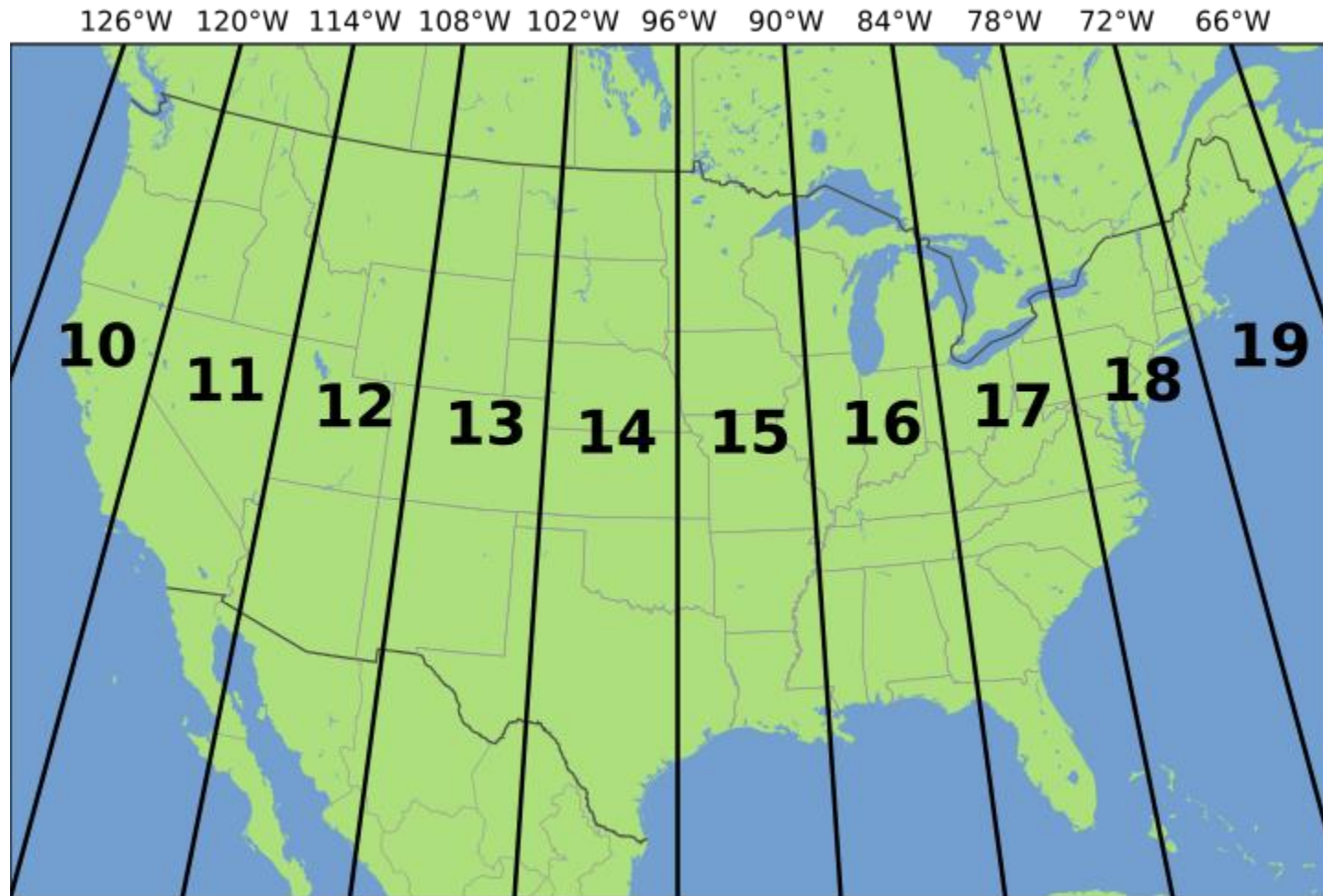
AERSURFACE CO pathway

- CO CENTERXY or CENTERLL
 - Location of center (i.e: surface met station)
 - CO CENTERXY *east north zone datum*
 - East & north: UTM coordinates
 - Zone: UTM zone
 - Datum: NAD27 or NAD83
 - CO CENTERLL *lat long datum*
 - Lat & long: no letter, **negative** long for western hemisphere
 - These coordinates not used in AERMET
 - E.g: CO CENTERLL 28.602304 -81.200204 NAD83

NAD & UTM

- Used to account for the curvature of earth
- NAD: North American Datum of 1927/1983
 - Datum: coordinate system defines the shape of earth
 - NAD 27/83 best suits north American
- UTM: Universal Transverse Mercator coordinate system
 - A simple algorithm to expand round earth to a flat surface, using a datum
 - Instead of uneven lat and lon, now you have a evenly distributed X and Y

UTM zone in US



AERSURFACE CO pathway

- CO DATAFILE
 - Provide names of NLCD land cover, % impervious and % tree data
 - CO DATAFILE *data type path filename*
 - *data_type*: NLCDXXXX: land cover; MPRVXXXX: % impervious; CNPYXXXX: tree cover
 - E.g.
 - CO DATAFILE NLCD2016 “NLCD.tiff”
 - CO DATAFILE MPRV2016 “NLCD_MPRV.tiff”
 - CO DATAFILE CNPY2016 “NLCD_CNPY.tiff”

AERSURFACE CO pathway

- CO CLIMATE
 - Set climate conditions
 - CO CLIMATE *sfc_mois snow_cover arid_condition*
 - *sfc_mois*: WET, DRY or AVG. Relative to normal climate conditions (30 y)
 - *snow_cover*: SNOW, or NOSNOW. Whether winter has snow cover
 - *arid_condition*: ARID or NONARID. Desert-like place? Used for NOSNOW
 - E.g.
 - CO CLIMATE AVERAGE NOSNOW NONARID

AERSURFACE CO pathway

- CO_FREQ_SECT
 - Set frequency & sectors (recall AERMET)
 - CO_FREQ_SECT *frequency #sectors airport_flag*
 - *frequency*: ANNUAL, SEASONAL, or MONTHLY. Season default winter to Dec, Jan, Feb, can be override using CO_SEASON
 - *airport_flag*: AP, NONAP, or VARYAP. Whether to use airport roughness values for all sector (AP), not use (NONAP), or vary by sector (VARYAP)
 - CO_SECTOR set each sector
 - *CO_SECTOR* *sector_index start_dir end_dir airport_flag*
- CO_RUNORNOT
 - RUN or NOT: Whether to start run or not

OU pathway

AERSURFACE OU pathway

- Start and end OU pathway
 - OU STARTING
 - OU FINISHED
- Between start and finish
 - OU SFCCHAR: output file name for AERMET
- Need CO DEBUGOPT to enable some debug files
 - OU NLCDGRID: land cover grid data
 - OU MPRVGRID: % impervious grid data
 - OU CNPYGRID: % tree canopy grid data